

# Problem with Imaginary numbers

## Problem

If  $A^x = B^y = C^z$ , where A, B, C, x, y and z are positive integers and x, y and z are all greater than 2, then A, B and C must have a common prime factor.

## Abstract

The factors of a number are, prime numbers (because of the fundamental theorem of arithmetic). but can also be composite of those. In this paper, we assumed that both A,B,C are prime, and conclude, that if they are prime, they don't have common prime factors. So, for example, we only test A=3 , and not A=2\*5 (A is not prime, in this case)

Examples:

1.  $A = 3$  ,  $x = 3$

1.1.  $A^x = 3 * 3 * 3 = 27$

We only consider, in this paper, the above.

2.  $A = 2*5$ ,  $x = 3$

2.1  $A^x = 2 * 5 * 2 * 5 * 2 * 5 = 1000$

Above, is not considered...

Another aspect, is that is only necessary to verify factors of A and B, B and C, or A and C, because if they haven't got a common factor, We don't need to go any further.

## Trying to prove

If A, B and C are prime, they don't share a common factor.

## Proof

**A prime, and B, prime. If they are different, there is no common prime factor.**

$$A^x + B^y = C^z \quad (1)$$

$A \neq B$ , and prime, so they have no common factor.

Example:  $2 * 2 * 2 + 3 * 3 * 3 = 35$ ...don't share common factor.

## Approach to problem

Methodology, is, choosing smaller number from y or x, and use that to put the factor (in this case A), in evidence. If  $y < x$ , put  $A^y$  in evidence, otherwise, put  $A^x$  in evidence. This is done, because, we are going to use exponents, and we don't want to use fractions.

It doesn't matter if  $x > y$  or  $y > x$ , because we are going to choose the smaller of them.

$$A^x + A^y = A^z \quad (2)$$

For  $x < y$ , we choose to put  $A^x$  in evidence, avoiding fractions

$$A^x(1 + A^{y-x}) = A^z \quad (3)$$

divide both side by  $A^x$

$$1 + A^{y-x} = \frac{A^z}{A^x} \quad (4)$$

Using equation 4, if  $x > z$ , we get negative exponent (the exponent is result of  $z-x$ ), so its a fraction. In that case we get

$$1 + A^{y-x} = \frac{1}{A^{something}} \quad (5)$$

This equation doesn't hold, because right side, can never be integer, and left side is integer.

If instead of  $x > z$  on equation 4 we use  $z > x$ , we get a positive exponent.

$$1 + A^{y-x} = A^{positive} \quad (6)$$

This equation doesn't hold because left side can never be on form  $A^{exponent}$  (the minimum value possible is 2, and must multiply the number, but is adding 1)

**For  $x=z$ , or  $y=z$**

on equation 4 If  $x = z$ , or  $y = z$ , we'll chose  $x=z$ , for explanation

$$1 + A^{y-x} = \frac{A^z}{A^x} \quad (7)$$

Doesn't hold, left side is bigger than one. right side is 1.

Other approach

$$A^x(1 + 1) = A^z \quad (8)$$

$$2 = \frac{A^z}{A^x} \quad (9)$$

Doesn't hold, left side 2, not equal to right side, 1.

## Using explicit numbers $x=5$ , $y=3$ , $z = 5$

Since  $y$  is smallest exponent, we put that on evidence

$$A^3(1 + A^{5-3}) = A^5 \quad (10)$$

$$1 + A^{5-3} = \frac{A^5}{A^3} \quad (11)$$

$$1 + A^2 = A^2 \quad (12)$$

Doesn't hold.

## Conclusion

**If A and B and C, are prime, they don't share a common factor**

## Other remarks

If we want to know the interval between  $3 * 3 * 3 + 5 * 5 * 5$ , number 152, factorize it in primes,  $2^3 * 19$ . So  $2 * 2 * 2 < 152 < 2 * 2 * 2 * 19$  (19 is prime, no factorization possible)..so, 152, can't be on form  $2 * 2 * 2 * 2 \dots$  or  $2^x$

We can also use parity on numbers...if even + even = even, they both share a two, meaning they share a prime factor. if odd + odd = even, They don't share a two. If odds are the same number, we have  $2 * \text{odd} = \text{even}$ , or  $\text{odd} = \frac{\text{even}}{2}$ .

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Bibliography

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